

Single-Axis Solar Tracker Project

Technical Documentation & Setup Guide

1. Project Overview

This project is an automated **Single-Axis Solar Tracking System** built using Arduino. The system is designed to orient a solar panel toward the strongest light source (the sun) automatically. This optimization increases the efficiency of solar energy harvesting by maintaining a perpendicular angle between the panel and the light rays.

2. Hardware Components

- **Arduino Uno:** The primary microcontroller board.
- **Solar Panel:** Captures light energy.
- **Servo Motor (SG90):** Responsible for tilting the panel left and right.
- **Light Dependent Resistors (LDRs):** Two sensors used to detect light intensity.
- **Resistors:** Used for voltage divider circuits with LDRs.
- **Breadboard & Jumper Wires:** For circuit assembly.
- **Structural Frame:** Wooden support for the panel and motor.

3. Wiring and Connections

A. Light Sensors (LDRs)

The LDRs are placed on opposite sides of the panel with a divider between them to create a shadow effect.

Component	Arduino Pin	Connection Type
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LDR 1 (Right)	A0	Analog Input (Voltage Divider)
LDR 2 (Left)	A1	Analog Input (Voltage Divider)

B. Servo Motor

The motor controls the horizontal axis movement.

- **Red Wire:** Connected to 5V.
- **Brown/Black Wire:** Connected to GND.
- **Orange Wire (Signal):** Connected to Digital Pin 9 (PWM).

4. Operation Logic

The system operates based on a comparative light intensity algorithm:

1. The Arduino reads light values from **A0** and **A1**.
2. It calculates the difference between the two values.
3. If the light on the right is stronger (beyond a set tolerance), the servo rotates to the right.
4. If the light on the left is stronger, the servo rotates to the left.
5. When the light is balanced, the motor stays stationary to save power and maintain stability.

5. Key Setup Observations

- **Shadow Divider:** The vertical black barrier between the LDRs is crucial for high-accuracy tracking.
- **Power Management:** The system is powered via USB or an external battery through the Arduino power jack.
- **Feedback Loop:** The real-time response shown in the video confirms that the code and wiring are correctly synchronized.

